REFLECTION REPORT

1. What did I initially expect?

I initially expected the attention model to show a clearer improvement over the simple forecasting baselines because the synthetic environment contains daily seasonality, weekly effects, trend, and rare spikes and drops. I also expected scaled attention to show a clearly better learning trajectory than unscaled attention. Finally, I expected attention to be most useful when information from several positions in the 24-hour history was relevant.

2. Which hypotheses were correct?

Hypothesis 1 was supported in the sense that the attention implementation successfully learned the controlled forecasting task and produced a usable warehouse forecasting model. Hypothesis 2 was partially supported: attention did not reduce MAE below persistence, but it did have lower RMSE and slightly lower MAPE. Hypothesis 3 was supported: the fixed model degraded under both the +40 level shift and the increase of noise standard deviation from 12 to 24.

3. Which hypotheses were wrong?

The strong version of Hypothesis 2 was wrong. I expected attention to clearly outperform simple baselines, but persistence had MAE 18.5331 versus attention MAE 18.5462 on the original test set. The expectation that scaled attention would necessarily finish with a lower toy-task loss was also not supported: at epoch 100, scaled attention had MSE 0.088134 while unscaled attention had 0.087712. Both learned substantially, but unscaled attention finished slightly lower in this run.

4. What surprised me?

The biggest surprise was how competitive persistence was. Because the synthetic process is strongly autocorrelated, simply using the latest observation already gives a strong forecast. A second surprise was that the scaling ablation did not produce a clearly lower final loss for scaled attention. A third was the size of the individual failures: the largest absolute error was 211.5344 demand units, with the largest errors associated with target jumps of roughly 189–199 units.

5. What was the hardest implementation problem?

I felt a bit harder to implement the training phase of Warehouse forecasting, which will manage all data with attention right! And I am also new to the concept of Gradient Verification, Numerical Stability, where I got to know and understand about it and later implementation.

6. What failure did I investigate?

I investigated the largest individual forecasting errors instead of stopping at aggregate metrics. For each test forecast I calculated absolute error and examined the previous demand and the target jump, defined as next-hour actual demand minus previous observed demand.

The five largest errors were 211.53, 210.74, 189.82, 184.82, and 182.47 demand units. Their target jumps were approximately 189.96, 199.09, 198.95, 188.77, and 189.10. This shows that abrupt one-hour changes are a major diagnostic pattern in the largest failures, although a target jump does not prove the exact cause of an individual error.

7. What do I learned?

Generally, as a student we all work on existing datasets and this is the first time I am working with the synthetic data that automatically generated every time with statistical properties. And I also understand about Gradient Verification and Numerical Stability. And I understand that generalization should be tested by changing the data-generating conditions while keeping the trained model fixed.

8. What remains uncertain?

I still do not know how much of this behavior would transfer to real warehouse data because the environment is synthetic. I also cannot interpret average attention weights as causal explanations. The two distribution shifts are limited to level and ordinary-noise changes; they do not test changes in seasonality, spike frequency, weekly behavior, missing observations, or other structural changes. Finally, the small d_model = 4, single-head model demonstrates the mechanism but does not establish how a larger attention architecture would behave.

9. Where did AI assistance help?

AI assistance helped as an implementation and clarifiying doubts partner. It was especially useful for repetitive engineering work such as checking tensor shapes, writing focused tests, and organizing experiment outputs. The experimental claims themselves were tied to the recorded project outputs rather than accepted from AI suggestions alone.

10. Where did I need to correct or reject AI-generated suggestions?

I had to keep the implementation constrained to the assignment instead of allowing a generic Transformer solution to replace the required first-principles attention. The implemented layer therefore directly performs Q/K/V projections, score calculation, scaling, stable softmax, and weighted aggregation. I also had to reject conclusions stronger than the evidence: the results do not justify saying attention simply beats the baselines, and the toy experiment does not justify saying scaled attention always gives lower loss. AI suggestions were treated as proposals to verify, not as facts.

11. If given one additional day, what would I investigate?

First, I would compare shorter and longer history windows to determine whether the model actually benefits from information beyond the most recent few hours. Second, I would generate matched datasets with and without spikes/drops to quantify their contribution to the RMSE behavior. Third, I would run multiple random seeds and report variability rather than relying on one training run. Fourth, I would test changes in spike probability, seasonal amplitude, and weekly effects. Finally, I would inspect attention patterns for individual successful and failed forecasts instead of only the average attention distribution.

12. what I would not claim

I would not claim that this experiment proves self-attention is superior to simple forecasting methods. I would not claim that attention weights are causal explanations. I would not claim that synthetic results establish real-world warehouse performance. I would not claim that scaled attention always produces lower loss. I would not claim that every large forecast error was definitely caused by an injected spike or drop. The strongest supported conclusion is narrower: the first-principles attention implementation was mathematically verified, learned the controlled toy task, produced competitive one-step forecasts on the designed synthetic warehouse environment, changed measurably under controlled distribution shifts, and exposed identifiable failure cases around abrupt demand changes.